Storage Units -- Guided Notes ANSWER KEY

Unit tp-1.5 | CompTIA Network+ N10-009 Obj. FC0-U71 1.3 | N10-008 FC0-U71 1.3

For instructor use / distribute after students complete the notes.

Question 1

The smallest unit of digital data is the _______, which holds a single binary value (0 or 1). Eight bits make one _______. A _______ stores 1,024 bytes, and a _______ stores 1,024 kilobytes.

Answer: Bit; byte; kilobyte (KB); megabyte (MB).

Question 2

Fill in the storage unit table:
1 KB = _______ bytes
1 MB = _______ KB
1 GB = _______ MB
1 TB = _______ GB

Answer: 1 KB = 1,024 bytes; 1 MB = 1,024 KB; 1 GB = 1,024 MB; 1 TB = 1,024 GB.

Question 3

Hard drive manufacturers advertise storage using the _______ (decimal/binary) definition where 1 GB = 1,000,000,000 bytes. Operating systems measure storage using the _______ (decimal/binary) definition where 1 GB = 1,073,741,824 bytes. This is why a 500 GB drive appears as approximately _______ GB in Windows File Explorer.

Answer: Decimal; binary; approximately 465 GB.

Question 4

A short text message is approximately 160 _______ (bits/bytes). A typical MP3 song file is around 4 to 5 _______. A 4K movie might be 50 to 100 _______. Choose the correct unit for each: bits, bytes, megabytes, gigabytes.

Answer: Bytes (characters); megabytes (MB); gigabytes (GB).

Question 5

_______ is the space used to run programs and store data while the computer is on (RAM). _______ is the space used to permanently save files, programs, and the OS (hard drive/SSD). The key difference is that memory is _______ (lost when power is off) while storage is _______.

Answer: Memory (RAM); storage (hard drive / SSD); volatile; non-volatile (persistent).

Question 6

A file is 750 MB. A flash drive has 1 GB of free space. Will the file fit? Show your reasoning:
1 GB = _______ MB
750 MB _______ (is/is not) less than 1,024 MB
Conclusion: the file _______ fit on the drive.

Answer: 1 GB = 1,024 MB; 750 MB is less than 1,024 MB; the file will fit on the drive.

Question 7

When choosing appropriate storage units, match each description to the correct unit:

A single character of text: ________

A Word document (5 pages): ________

A smartphone photo: ________

A Blu-ray movie: ________

Answer: A character: 1 byte; Word document: approximately 30-50 KB; smartphone photo: 3-8 MB; Blu-ray movie: 25-50 GB.

Question 8 -- Real World Application

REAL WORLD: A student buys an external hard drive advertised as 2 TB. When they plug it in, Windows shows only 1.81 TB of usable space. The student believes they were sold a defective or undersized drive and wants to return it. Explain what is actually happening, using the decimal vs. binary definition of gigabyte, and whether the drive is defective.

Answer: The drive is not defective -- this is expected behavior caused by the difference between decimal and binary storage definitions. The manufacturer defines 2 TB as 2,000,000,000,000 bytes (decimal, powers of 10). Windows measures storage in binary units where 1 TB = 1,099,511,627,776 bytes. Dividing 2,000,000,000,000 by 1,099,511,627,776 gives approximately 1.818 TB as shown by Windows. No data capacity has been lost -- it is simply a unit definition difference. The student should not return the drive.